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Rhetoric versus Reality

A Review of Studies on State Enterprise Zone Programs

Margaret G. Wilder and Barry M. Rubin

This paper provides a comprehensive review of empirical analyses of state-sponsored enterprise zone programs. An introductory segment focuses on major theoretical and policy issues about enterprise zones. This discussion is followed by a summary of studies on two major dimensions of enterprise zone programs: (1) job and investment impacts, and (2) program costs. The research reviewed ranges from individual case studies to cross-state analyses. Interpretation of findings from these studies suggests that enterprise zones vary in their effects on investment and employment in declining areas. This variation is partly explained by the features of program design and the specific attributes of zone contexts. Although enterprise zones have been effective in generating new employment and investment in certain areas, they have important limitations that must be scrutinized more closely to minimize program costs and target benefits to depressed communities more effectively.

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The policy debate about enterprise zones has failed to reflect the insights derived from a growing body of empirical assessments of state-initiated programs. In many respects, this debate is mired in the concept of enterprise zones that the Reagan and Bush administrations envisioned and promoted. It is seldom recognized that while efforts to establish a federal enterprise zone program failed until the Clinton Administration embraced the idea in the form of "empowerment zones," a majority of state governments had forged ahead during the 1980s, adapting the zone concept to their particular economic development agendas.[1] As of 1995, 34 states had active enterprise zone programs,(See table 1.) With the enactment of the new Federal Empowerment Zone and Enterprise Community Program, the enterprise zone concept has been extended to 106 additional communities.[2] Notwithstanding the potential importance of Federal initiatives, enterprise zones are already a widely utilized tool in state economic development (Green 1990). Evaluations of the Federal initiative are only beginning to take shape, but a wealth of insight can be gleaned now from existing assessments of state experiences with enterprise zones.

The purpose of this review is to inform the ongoing policy debate and program efforts by examining findings and conclusions drawn from empirical studies of state-initiated enterprise zone programs. Although the primary objective of this paper is to synthesize insights derived from previous studies, the implications of those research efforts must be drawn as well, with reference to the conceptual and theoretical issues raised by the development of enterprise zone programs. Thus we begin with a summary of the policy debate and the critique of enterprise zones. Next, we provide a review of research studies on two major aspects of enterprise zone programs: their effects on jobs and investments, and their costs. The final segment of the paper presents a set of planning and policy implications.

As the following review reveals, program experience has made two

Table 1. State enterprise zone programs

State	Year Began	Number of Zones
Alabama	1988	12
Arizona	1988	11
Arkansas	1984	458
California	1986	25[1]
Colorado	1986	16
Connecticut	1982	11
Delaware	1985	30
Dist. of Columbia	1988	3
Florida	1981	30
Georgia	1982	3[2]
Hawaii	1987	pending (24 max)
Illinois	1983	90
Indiana	1983	15
Kansas	1982	74
Kentucky	1982	10
Louisiana	1981	1625
Maine	1987	program ended
Maryland	1982	17
Michigan	1986	1
Minnesota	1983	5
Mississippi	1984	program ended
Missouri	1983	50
Nebraska	1992	2
Nevada	1984	2
New Jersey	1984	10
New York	1987	19
Ohio	1982	227
Oklahoma	1983	88
Oregon	1986	30
Pennsylvania	1983	45
Rhode Island	1991	9
South Carolina	1987	3
Tennessee	1984	2
Texas	1988	103
Utah	1988	15
Vermont	1986	program ended
Virginia	1984	18
West Virginia	1987	program ended
Wisconsin	1988	8

Source: Derived from data reported in "State Enterprise Zone Update," prepared by the National Association of State Development Agencies (1995), and informal HUD data summaries

[1]California also has designated nine "economic/employment incentive areas" that are economically distressed. However, firms are required to meet certain hiring guidelines to receive program benefits.

[2]All three Georgia zones are located in the city of Atlanta.

realities clear: enterprise zones have not lived up to the broad panacea-like imagery conjured up by early Federal proposals and rhetoric; neither, however, have they brought about the massive dislocations and dominance of large industries predicted by zone critics. The answer to the enterprise zone riddle is far more complex than either its promoters or detractors suggest. More importantly, the base of knowledge drawn from the states' experiences with enterprise zones suggests both the limitations and the potential utility of this economic development tool.

One weakness in the base of knowledge about state enterprise zones is the lack of attention to the planning process as an element in program implementation. The failure to include planning process requirements in state legislation that authorizes enterprise zones may critically influence program outcomes; yet this issue is generally not addressed in major studies of the effects of enterprise zones. The omission of planning elements from state enabling legislation for enterprise zones and from subsequent program evaluations should be of special interest to planners. The implications of these omissions are discussed in more detail in the final portion of the paper.

The Theoretical and Policy Debate

The core of the policy debate over enterprise zones is whether development incentives are effective in arresting and reversing urban decline.[3] Supporters of the concept argue that such development incentives are necessary to leverage investment in areas perceived as poor risks (Hall 1977, 1982; Butler 1981, 1992; U.S. Chamber of Commerce 1981; Kemp 1982, 1989, 1992; Weiner 1984; Congressional Digest 1985; Cowden 1992; U.S. Congress, House of Representatives 1992).[4] Other advocates, such as the National League of Cities, support enterprise zones in concept, but argue that tax incentives must be accompanied by direct capital investment in infrastructure and public services in order to increase the appeal of urban areas to potential investors (U. S. Congress, House Committee on Banking, 1992). Such investment not only stabilizes a declining area, but is seen as having positive multiplier effects that spill over to neighboring areas.

Other commentators have vehemently questioned the viability of the enterprise zone approach. Some researchers argue that incentives such as tax abatements and regulatory relief have little or no influence on investment or business location decisions (Harrison and Kanter 1978; Peirce, Hagstrom and Steinbach 1979; Vaughn 1979; Jacobs and Wasylenko 1981; Schmenner 1982; Hawkins 1984; Rubin and Zorn 1985). Critics argue that the modest savings provided by development incentives are often too small a proportion of business costs to affect such decisions. Specifically, enterprise zones tend to benefit primarily large, capital-intensive firms, which have higher tax liabilities, but provide little assistance to small or labor-intensive firms, which have more potential for employment growth (Jacobs and Wasylenko 1981; Clarke 1982; Mier 1982; Walton 1982; Glickman 1984; Hawkins 1984). Moreover, those jobs created by zone businesses are much more likely to be low-paying and provide little job security (Clarke 1982; Goldsmith 1982).[5]

A slightly different line of critique focuses on the assumed direct and indirect costs associated with enterprise zones. It is argued that their incentives in the form of forgone tax revenue and public capital expenditures increase the tax burdens for nonzone firms and residents (Armstrong 1981; Humberger 1981; Mounts 1981; Clarke 1982; Hawkins 1984). Moreover, the zones tend only to transfer capital investment from one location to another, rather than generating "new" business activity (Peirce, Hagstrom and Steinbach 1979; Clarke 1982; Massey 1982). Finally, some observers argue that the opportunity costs of enterprise zones reduce direct aid to urban communities, and that more attention and resources should be focused on programs such as education and job training (Birdsong 1989; Estes and Hamond 1992; Levitan and Miller 1992).

The debate over enterprise zones has raised a number of issues about their effectiveness. Most of the debate centers on two key questions: (1) Do the development incentives provided by enterprise zones generate new investment and employment in declining urban areas? (2) What direct and indirect costs or burdens are attributable to enterprise zones? Although neither of these questions has been fully answered, significant attempts have been made to describe and analyze the nature of zone programs and their economic effects. The appendix lists key attributes of major studies to date. We review the research results that assess: (1) effects on investment and employment, and (2) program costs.

Studies of Effects on Investment and Employment

Although the economic development literature often discusses the *potential* effects of enterprise zones, empirical research on, or analysis of zone programs is somewhat limited. The modest amount of empirical research is due to two basic constraints: (1) the lack of reliable quantitative data to evaluate zone performance, and (2) the difficulty of isolating the effects of zone designation and incentives from those of other economic development factors and initiatives. Nevertheless, a small body of empirical research does exist and continues to grow.

Descriptive Studies

The first direct study of an enterprise zone's economic effects was carried out by Earl Jones (1985) in a case study of Bridgeport, Connecticut. Jones compared the number of property transfers and building permits issued within the zone to those in a similar, but nonzone area. He found only marginal differences in transfer and building activities between the zone and nonzone areas.

A few years later, Jones (1987) undertook a similar study for Decatur, Illinois. His analysis of data for a five-year period revealed significant increases in both the number and the value of building permits issued after zone designation. Jones expressed skepticism about the significance of these findings, however, concluding that the observed differences in building activity could be attributed to factors other than enterprise zone designation (i.e., a general upswing in building activity). He argued that the changes might very well have occurred even in the absence of the enterprise zone program.

It is important to note, however, that the type of data—property transfers and building permits—used by Jones have certain limitations for analyses of zone effects, since they do not include such forms of new investment or employment as capital equipment purchases or added work-shifts. These initial studies show the difficulties of ensuring that the data to measure zone effects are appropriate, and the complexity of separating the effects of zone programs from those of other economic factors. These issues have been addressed both directly and indirectly in subsequent studies.

Ten-Case Study by HUD. One of the most comprehensive studies of enterprise zone impacts was carried out by the U.S. Department of Housing and Urban Development (1986). HUD assessed ten enterprise zone programs in nine states,[6] using data derived from a series of interviews with approximately 200 state and local officials, as well as with business and neighborhood representatives. The survey results showed that 263 zone firms were responsible for over $147 million in new investment, and 7,348 new or retained jobs.[7]

The types of zone firms varied in their capacity to generate jobs. New business start-ups accounted for approximately 33 percent of new jobs, and existing firms for 27 percent. Only a small proportion (14 percent) of new jobs was generated by relocating firms. New branch expansions of outside firms created another 12 percent of the employment growth.

The targeting of new jobs to disadvantaged residents varied across the case study sites. The proportion of new (or retained) jobs held by "low and moderate income persons" ranged from 13 percent in Michigan City, Indiana to 90 percent in Thief River Falls, Minnesota. The proportions of new (or retained) jobs held by zone residents were somewhat lower, ranging from 5 percent in York, Pennsylvania to 70 percent in Bridgeport, Connecticut, and Chicago.

More detailed analysis revealed that most of the new investment (64 percent) in enterprise zones was made by either new business start-ups or the expansion of existing zone firms. New branch operations generated 20 percent of investment growth, and relocating firms 11 percent. Of the new investment, most was made by small firms; approximately 63 percent of the firms making new investments in the zone had fewer than 50 employees. Only 4 percent of investing firms had more than 200 workers.

In addition to these insights, the HUD study revealed some of the subtler, but important aspects of zone programs. For example, despite significant new investment and job growth, some zones had net job loss. The study also showed that zones varied dramatically in their rates and patterns of new investment. Although some zones had dramatic and steady growth in investment, others had initial growth followed by a noticeable slowdown. Other zones remained stagnant and continued to decline economically. The HUD study acknowledged the difficulty of obtaining reliable data on zone effects, and cautioned that the observed outcomes for jobs and investment could not be attributed solely to the enterprise zone programs.

The HUD study also assessed the perceptions and attitudes of program participants about the effects of zone designation and incentives. Respondents indicated that zone designation had encouraged significant business investment. Interestingly, the zone designation itself was found to be more important in inducing this investment than the tax incentives themselves were. Incentives did not play a major role in most firms' location decisions, although some firms were partly influenced by them. Existing zone firms, in contrast, viewed incentives as far more influential in their expansion and investment decisions.

While the HUD study is useful in delineating major characteristics of enterprise zones, its findings must be qualified, because it used interviews with zone officials, who are potentially biased by a natural tendency to attribute any job growth within the zone exclusively to the enterprise zone's influence. Such officials tend to ignore job losses, focusing on gross rather then net job growth. Moreover, they often provide highly inflated estimates of job retention, which is admittedly very difficult to measure. The issue of data bias and/or reliability can be raised with many of the studies reviewed here that rely on interviews and perceptions rather than verifiable data.

California Program Assessments. An initial assessment of the California Enterprise Zone Program (implemented in 1987) was conducted by the Office of the State Auditor General (1988). By using a sample survey of the firms in the ten zones, rather than interviews with zone officials, this study employed a more reliable methodology than that in the HUD study. The Auditor's report concluded that economic activity had increased somewhat in the state's original ten enterprise zones and three "incentive areas."[8] An estimated 3,600 new jobs had been generated by firms within the zones and incentive areas. The survey results showed net employment gains ranging from 4.7 percent in the Los Angeles-Central City zone to a high of 61.2 percent in the Eureka zone. The survey sample firms alone reported a net increase of over 1,800 employees. Most significantly, 1,151 jobs within the zones went to economically disadvantaged persons. Despite these positive outcomes, however, the zones were not insulated from job losses. Since the zone program had begun, over 2,100 zone firms had closed or relocated. The losses were counterbalanced by a gain of over 2,200 new businesses.

A subsequent study of zone firms by Litster (1990) directly challenged the accuracy of the Auditor's report. Analyzing responses to a survey of zone businesses, Litster found that 2,518 (rather than 3,600) new jobs had been created in the California zones. The majority (54 percent) were created by new businesses. The balance of new employment was almost equally divided between existing firms that had expanded operations and those that had relocated from outside the zones.

Litster was particularly interested in the impact of the California program on the locational decisions of new and expanded zone firms. A survey of 137 participating businesses revealed that only 24 percent viewed enterprise zone designation/incentives as a strong factor in their decisions about location. About 55 percent of the firms placed "little or no importance" on zone designation or incentives in their location decisions. Location decisions were overwhelmingly influenced by the factors of real estate costs, site characteristics, road/rail access, and proximity of product market. Although tax credit incentives did not play a primary role in their location decisions, 40 percent of the survey respondents had used at least one of those available.

A more recent evaluation of the California zone program by Dowall, Beyeler, and Wong (1994) used County Business Patterns data on employment and establishments to estimate job generation and business development. These researchers found job growth across most of the zones and incentive areas. Between 1986 and 1990, employment had increased in all of the targeted areas except the Los Angeles-Watts incentive area (which suffered a 2 percent net job loss). Net employment growth ranged from 30 jobs in the Porterville zone to 7,400 jobs in the San Jose zone. About 54 percent of employment growth was generated by large firms (i.e., those employing 100 or more persons). Small firms (i.e., those with fewer than 20 employees) were responsible for 13 percent of job growth; medium-sized firms generated 33 percent. Net business growth occurred in 9 of the 13 targeted areas. The San Jose and Los Angeles- Central zones were leading growth centers, with 140 and 80 net new businesses, respectively. The Fresno zone had the only significant decline, with a net loss of 100 establishments.[9]

In an effort to identify the underlying causes of the employment and business growth, Dowall, Beyeler, and Wong conducted a shift-share analysis to isolate the effects of the zone/incentive area program.[10] Their findings suggest that five of the zone/incentive areas[11] were at a significant competitive disadvantage compared to their surrounding county and regional areas. However, three of the targeted areas (San Jose, Los Angeles-Pacoima, and Sacramento-Northgate) seemed to have had a competitive advantage. The remaining targeted areas showed only modest zone effects. The researchers posited that the zone/incentive area initiatives were not conclusively associated with any of the positive outcomes. A phone survey of business owners operating in the zone/incentive areas revealed that about one-half were actively taking advantage of the incentive program. Among program participants, about 33 percent reported that the program had influenced their decisions about location or expansion, and 20 percent reported it had influenced them to expand their workforce.

It is worth noting that this study introduced a new data source (i.e., County Business Patterns zip code data). Although this database is more reliable than interviews with zone officials, there are inherent difficulties with disaggregating the zip-code-level data for the distinct subareas encompassed by enterprise zones. The researchers apportioned job and establishment data according to the proportion of a zone's land area represented by the zip code. Due to the potential biases embedded in this approach (which the researchers acknowledged), job and business development may have been over- or under-estimated.

The Maryland Program. The U.S. General Accounting Office conducted a quantitative assessment of the effects of three of the Maryland enterprise zones (U.S. General Accounting Office 1988, 1989). GAO researchers applied interrupted time series analysis to employment data for the Hagerstown, Cumberland, and Salisbury zones. The analyses revealed that both abrupt and gradual increases in employment had occurred after the enterprise zones took effect. At each site, the employment growth contrasted sharply with job trends prior to designation. Both employment and investment were heavily concentrated in existing firms.

Interviews with relevant employers within the zones, however, suggested that the enterprise zone program did not necessarily account for the changes. Rather, a fairly complex relationship between program incentives and business decisions was apparent. In general, firm representatives said that tax incentives played little or no role in their locational decisions, which were far more affected by market conditions, site characteristics, and general community attributes.

However, the survey revealed that incentives did have a very strong effect on hiring and investment decisions. About 87 percent of the firms surveyed regarded the program's property tax incentive as important to their operations. Similarly, three out of four firms indicated that the program's "new hire" credits had influenced their hiring decisions. Interestingly, attitudes about program incentives showed little variation across industrial categories or firm sizes. The study concluded that the zone program had clearly influenced some firm behavior, but no clear causal link exists between job growth and zone designation or incentives.

Although the findings are useful, this study must be viewed with caution due to the nonrandom sample of zones: all are semi-rural or rural areas. (Rural communities have decidedly different employment dynamics from those of urban areas.) Another concern is that only 13 percent of the study's survey respondents had participated in the enterprise zone program. There is clearly a potential bias in responses from program participants as compared to nonparticipants.

The New Jersey Program. An assessment of the New Jersey Urban Enterprise Zone Program conducted by Marilyn Rubin and Regina Armstrong (1989) concluded that the program had indeed improved the "relative economic position" of the state's most distressed cities. A survey of 976 firms in the state's ten enterprise zones indicated significant growth in jobs and investment. According to the study, an additional 9,193 jobs and $803 million in private investment had accrued in the enterprise zones since the program's inception in 1985.[12]

The study also examined the influence of program incentives on the behavior of zone firms. For 32 percent of the firms, zone incentives were cited as a primary reason for location or expansion decisions; they were a secondary factor for 38 percent. However, the remaining firms (30 percent) were not influenced by zone incentives. In general, the influence of zone incentives increased with firm size; those with 50 or more employees were much more likely to cite zone incentives as a primary reason for their location and investment decisions.

The New York Program. The initial evaluation of the New York Enterprise Zone Program was completed in 1990, just three years after the first cohort of zones was designated,[13] by the consulting firm of Hamilton, Rabinovitz, and Alschuler (1990). Data drawn from state "business activity reports" showed that approximately 1,200 new jobs had been created in the state's enterprise zones between 1988 and 1989. The Brooklyn and Syracuse zones led in terms of employment growth, with 500 and 300 new jobs, respectively. A survey of zone coordinators and business owners, and detailed case studies of the East Brooklyn, Olean, Troy, and Syracuse zones revealed more subtle effects. The majority of participating firms had previously been located either within a zone (47 percent) or within New York State (18 percent). Locational decisions by zone firms were influenced by three primary factors: familiarity with the area (59 percent of firms), availability of zone program tax incentives (54 percent), and proximity to customers (48 percent).[14] Approximately 42 percent of zone firms felt that the program had directly affected investment decisions, and 14 percent reported a marginal effect. In contrast, 43 percent of zone firms reported that they were not influenced by the program. Although the state program benefits influenced some new investment, firms within the New York City zones found local incentives even more compelling. Job development was less directly influenced by program incentives. Only one-fourth of the participating firms reported that the program influenced their employment levels, and one-third reported that the scale of their workforce was unrelated to the program.

Analytic Studies of Causal Links

As much of the preceding review has shown, enterprise zones have been associated with variable levels of new investment and job growth. However, the exact nature of the assumed relationships has not been adequately established. Two important unresolved issues are raised: (1) What factors explain the variations in zone effects? (2) Can observed changes in economic activity be directly attributed to enterprise zones? A study by Sheldon and Elling (1989) sought to address the first of these questions. The second question has been addressed in studies by Rubin and Wilder (1989), and Erickson et al. (1989). A brief review of the findings from these studies is presented below.

Sheldon and Elling's Midwest Study. In an effort to explain the variability in zone program outcomes, Sheldon and Elling (1989) studied the economic attributes and effects of 47 enterprise zones in Illinois, Indiana, Kentucky, and Ohio. Although each of these states had succeeded in attracting some new investment and employment to at least some of its zones, the researchers found a high degree of *intra-state* variability. Within each state there were examples of zones that had exceeded program objectives as well as of zones that had failed to meet them. Employment growth across zones ranged from zero to 3,076 new jobs, with an overall mean value of 499 jobs and a median of 225 jobs. Comparable numbers for "retained" jobs ranged from zero to 10,890 overall, with a mean of 868 and a median of 150.

The study also examined the nature of zone firms that had received development incentives. In each state, the majority of participating firms (57–70 percent) were existing businesses that had expanded. Many of the remainder (22–31 percent) were new start-up firms. The proportion of participating firms that had relocated to take advantage of incentives ranged from 19 percent in Kentucky to a modest 6 percent in Ohio.

Sheldon and Elling concluded that variations in investment and job generation across zones were most likely to be due to differences in: (1) local economic conditions (e.g., costs and condition of transportation, labor conditions, tax rates); (2) state and regional market conditions; and (3) state economic development policies. Moreover, they observed that the most successful zones were those in which "pure" zone strategies (i.e., tax incentives) were complemented by more traditional supports for economic development (e.g., technical assistance, location/site analysis, special staffing).

The Sheldon and Elling analysis is useful in sorting out the variables that may affect business investment and location. What is less clear is how these variables affect job generation. That question has been examined in two other studies.

Wilder and Rubin's Indiana Study. Wilder and Rubin (1988) undertook a detailed case study of the Evansville, Indiana enterprise zone. Using firm-level survey data, they found that from 1983 to 1986 (the first three years of program operation), 47 new businesses had been created in the zone, and 14 existing businesses had expanded.

Of the 1,651 new full-time jobs created, approximately 30 percent went to zone residents. Subsequent studies of Indiana's zones by Papke (1988, 1989) reported that residents of the Evansville zone obtained about 43 percent of new zone jobs in 1986, and 14 percent in 1987. In addition, another 1,000 new jobs were estimated to be retained, although this figure could not be verified.

The Evansville study partly confirmed criticism of zone programs, in that much of the new employment was generated by new branch operations of large firms. However, contrary to predictions by zone critics, the majority of the new businesses had not relocated from outside the zone.[15]

Wilder and Rubin concluded that the relative success of the Evansville zone was due to: (1) the economic viability of the zone area, (2) a high concentration of industrial land use, (3) autonomy of the zone's administration from local government, (4) the credibility and competency of the zone manager, and (5) the community organization role played by the zone association.

In a follow-up analysis of the Evansville zone, Rubin and Wilder (1989) sought to determine how much of the new job development was attributable to the enterprise zone program. Differentiating job development due to zone designation or incentives from that due to other influences is a methodological problem that has plagued studies of enterprise zone performance. To overcome the problem, Rubin and Wilder applied a shift-share analysis, using firm-level data reported for state income tax purposes. This methodology isolated the economic effects of the enterprise zone from those of the surrounding regional and national economies. The analysis showed that between 1983 and 1986, a net increase of 1,878 jobs occurred in the Evansville zone.[16] Of this total, 325 jobs resulted from general economic growth in the metro area, and 123 jobs from the zone's industrial composition. The remaining 1,430 jobs were attributed to the comparative advantage of the enterprise zone.

Although Rubin and Wilder stopped short of asserting a cause and effect relationship between the zone and job growth, they argued that the contrast between the distress of the zone area prior to designation, and its post-designation growth and vitality is most logically linked to the zone's operation. One of their most interesting findings was that there was no direct relationship between tax benefits to an employment sector and the extent of employment growth in that sector. In fact, they observed that the majority of tax credits went to Evansville firms that created no new employment. They cautioned that this finding should not be taken as indicating that the incentives were ineffective, since firms with no new employment might well have retained jobs as a result of zone incentives. They concluded, however, that zone programs such as Indiana's should be revised to link tax breaks more directly to job development.

Erickson, Friedman, and McCluskey's Cross-Sectional Analysis. In a more comprehensive study, Erickson, Friedman, and McCluskey (1989) analyzed HUD data for 357 enterprise zones in 17 states, and 2,014 business establishments. The authors examined the relationship between the structural characteristics of enterprise zone programs (e.g., number/type of incentives, number/size of zones) and the observed changes in employment and investment.

Their findings, contrary to predictions by critics of enterprise zones, revealed that only 7.5 percent of new zone businesses were branch operations of larger firms. Moreover, only 9 percent of the firms making new investments within the zones had relocated from outside the zones. In fact, about 55 percent of firms making new investments were existing ones that were expanding, and 26 percent were new firms. Smaller firms (i.e., fewer than 50 employees) and those in manufacturing, retail, and services dominated the new investment. The largest investments were made by "retained" firms (i.e., whose closure or contraction had been averted); they spent an average of $8.6 million to maintain or improve their operations.

In terms of job growth, the zones studied had an average growth of 464 jobs (since their respective designations), with a median of 175 new positions.[17] These jobs were created over, generally, a two-year period. The average gross job gain of 232 jobs per year represents small to modest growth; average employment at the time of zone designation was 4,776 jobs. Analyzing a subgroup of 66 zones for which baseline employment data were available, the researchers found that almost one-third of these zones had levels of gross job growth above those for the nation during the study period. Since the zones were severely distressed, with unemployment, poverty, and economic stagnation, the researchers concluded that a significant proportion of these gross job gains can be attributed to the operations of the enterprise zones. However, the median of 175 jobs indicates a substantial variation in their ability to generate or retain jobs.

(As Erickson et al. note, these numbers must be interpreted bearing in mind the relatively short period of time that enterprise zones have been in operation, and their typically small geographic areas). For the study sample as a whole, the authors found that, on average, 61 percent of new jobs went to zone residents, 52 percent to low-income persons, and 48 percent to the unemployed.

In an effort to explain variations in job and investment growth across zones, the researchers conducted a regression analysis of data for a subset of zones. The regression analysis revealed that 36 percent of the variation (i.e., increase in R^2) in business investment across zones could be explained by four variables: zone population, zone size, frostbelt location, and metropolitan location. Each of these variables was positively related to investment activity in zones. Two additional variables—minority population level and industrial land use—were also positively related to additional development, each accounting for 8 percent of the variation in investment.

A separate regression equation revealed the effects of state policy on investment levels: in combination, state policy variables accounted for 45 percent of the variation. A strong positive relationship was observed between the number of incentives provided to zone businesses and investment by those businesses. In contrast, an equally strong negative relationship existed between the number of enterprise zones in a state program and the respective zone investment levels.

Similar regression equations were derived to explain variations in employment growth, but their explanatory power was less satisfactory, explaining only 32 percent of the variation. Employment growth was positively related to zone population size, percentage of minority population, and proportion of industrial land use. In terms of state program attributes, the regression model revealed a strong positive relationship between the number of business incentives and job growth.

To complement these findings, the researchers interviewed the coordinators of 21 "high performance zones."[18] From survey responses, the researchers identified four common explanations for zone success:

1. The zone encompassed an area that was still economically viable.
2. Zone designation served as a catalyst and/or stabilizer, but was not the sole determining factor in revitalization.
3. The number and variety of zone incentives added to the programs' effectiveness. However, tax incentives (e.g., for property, inventory, and sales) were the most frequently used.
4. Strong local support from the private and public sectors increased effectiveness.

Although the researchers concluded that their analysis did not establish a clear causal relationship between enterprise zones and observed economic growth, they felt that considerable evidence pointed to a "significant marginal and catalytic role" (Erickson et al. 1989, 103).[19] While the results obtained by Erickson et. al. imply that some zones are successful, two caveats about their methodology are in order. The first is that their data were obtained from zone coordinators, who may have a tendency to overestimate the number of jobs created or retained. Second, their employment data represent gross, rather than net job growth. Without information on job losses in the zones as well, it is difficult to assess their overall effects on job growth and retention.

Interpretation of Investment and Employment Studies

Many of the issues in the theoretical and policy debates over enterprise zones can be directly assessed in light of this empirical review. The critique of enterprise zones has focused on: (1) general effectiveness, (2) the influence of development incentives, (3) displacement effects, and (4) job quality and stability. The studies reviewed here provide a basis for some initial answers.

The General Effectiveness of Enterprise Zones

The most consistent finding in studies of enterprise zones is that of increased job growth and investment. In most studies, there is a clear temporal relationship between the establishment of an enterprise zone program and increased economic activity within certain zoned areas. There is also an apparent targeted effect, in that the zoned areas frequently show economic activity that is not characteristic of either comparable subareas or the surrounding metropolitan area. Thus, it appears that the unique aspect of enterprise zone programs, geographic targeting, has been somewhat realized.

Despite that generally positive aspect, the evidence is equally clear that in certain urban areas, enterprise zones have failed to generate significant economic growth. Every study that examined data from multiple cases revealed variable outcomes. Variability in job growth and investment was found between state programs, as well as between zones within the same state. In some zones, job losses easily outnumbered marginal increases. This mixed picture of zone effects points to their limitations. Enterprise zones do not have magical qualities that can overcome all physical, social, and economic barriers to revitalization. In this respect, zone critics are correct in arguing that the myriad social and physical problems plaguing many urban neighborhoods (e.g. decaying infrastructure, high crime rates, inadequate school systems) are not responsive to targeted development incentives.

Enterprise zones do appear able to stimulate new employment and investment in certain areas. However, their use requires attention to program design features that can increase the probability of desired outcomes. Specific suggestions for policy development are presented below.

Job Access, Quality and Stability

Regardless of the context, the primary objectives of enterprise zone programs are to create jobs and attract private sector investment in declining areas. Not surprisingly, job growth has been the most frequently used measure of zone success. As the preceding discussion has shown, job creation has varied across zones, and some zones have both gained and lost jobs. Most studies of enterprise zones, however, do not report job losses.

Three tendencies are observed in zone job growth. First, most new jobs are generated by existing firms and/or new business start-ups. Second, new employment is heavily concentrated in manufacturing and wholesale/retail trade. Third, most new jobs are within firms with fewer than 50 employees. Although these observable outcomes are important, the picture of job creation is far from complete.

Two major concerns about job creation in enterprise zones have yet to be explored. First, job quality: critics have warned that jobs created in enterprise zones would tend to pay low wages and have little security. Unfortunately, most zone studies have not examined the skill levels, wages, or longevity of the jobs created. Second, the extent to which the new jobs benefit zone residents is not clear. In the few studies examining the employment of zone residents, the findings show great variation across zones, with roughly 20 to 30 percent, on average, of new jobs going to zone residents. However, the respective percentages for individual zones range from less than 2 percent to above 90 percent. These variations are observed between zones within the same state, as well as between states. In Indiana's zones (perhaps the most well-studied), variations in the percentage of jobs going to residents have occurred over time, as well. Between 1983 and 1986, the proportion of new jobs taken by residents of Indiana's enterprise zones ranged from less than 5 percent to more than 70 percent (Wilder and Rubin 1988). Subsequent studies found comparable figures, ranging from 1.4 to 100 percent in 1986, and from 5.7 to 37.2 percent in 1987 (Papke 1988, 1989).

To date, no specific factors have been identified that explain this variability. What is apparent is that most state programs do not tie major incentives directly to targeted employment. Instead, they tend to provide tax credits for employing zone residents and/or other disadvantaged persons—credits so modest that studies report the zone businesses view them as "not worth the trouble." The opportunity to directly target employment is thus seriously compromised.

If zone residents are not necessarily benefiting from job growth in certain zones, who is? A few studies have examined this question and report that non-zone residents receive many of the new jobs. Overall, zones have apparently created new jobs, but as a targeted employment strategy they have an uneven record.

Influence of Development Incentives

Much of the debate over enterprise zones focuses on their assumed power to attract new investment with development incentives. The picture that emerges from enterprise zone studies suggests that the relationship between incentives and development is complex. Virtually every study has found that employment and investment activity increased in *most* zones after program implementation. However, there was seldom a simple or direct relationship between the new economic activity and the development incentives. Some studies found that many qualified zone firms had not taken advantage of incentives, yet had engaged in new or expanded activity. This outcome, some researchers argue, reflects a more indirect effect of enterprise zones: that is, that the business community takes the mere existence of the zone program as a signal that the city and state have made a commitment to the zone area. That perceived commitment has a positive influence on decisions about investment and location. In such instances, enterprise zones have a kind of policy "placebo" effect. Hence state and local decision-makers may support enterprise zones as economic development policy because the zones are a useful public relations and marketing device (Wolkoff 1992).

Other studies suggest that incentives influence business location decisions primarily at the margins. Study interviews with business owners/managers reveal a remarkable consistency in their attitudes toward development incentives. Traditionally recognized location factors such as proximity to markets and transportation access were consistently acknowledged as more critical than development incentives in site selections. However, enterprise zone incentives were significant when the more central factors for competing locations were equal. The program incentives then became a "tipping point" that favored zone areas.

Zone studies also show that certain types of firms are much more likely than others to view development incentives as critical to their investment decisions. One of the most consistent findings is that existing zone firms are more likely than newcomers to take advantage of incentives provided. They typically undertake expansion facilitated by tax breaks. Similarly, incentives are favored by larger firms (i.e., over 50 employees), with more capital assets, who are often attracted by inventory-related and capital investment credits.

The debate over development incentives has been framed primarily as a "do or don't" proposition. Moreover, much of the research that fuels this debate has focused on the influence of incentives on decisions about site locations. A recent review of research on development policies by Bartik (1991), for example, suggests that at least in the case of taxes, incentives can have a positive impact on development activity. Enterprise zone studies add an important dimension to this discussion, pointing out that development incentives can play at least two distinct roles. In *location* choices of new and relocating firms, incentives play only a marginal role, but *investment decisions* by existing zone firms are heavily influenced by development incentives. Moreover, regardless of firm type, development incentives play a symbolic role; they seem to serve as an important index of business climate and commitment by the public sector.

Although development incentives appear to positively influence investment decisions of existing firms, no *direct* link between incentives and job creation has been shown. Zone research suggests a more subtle relationship, however. Most new jobs in enterprise zones are in existing or start-up firms. Since these types of firms are the most frequent users of development incentives, one can posit an indirect relationship between incentives and job creation.

However, caution must be exercised in equating business development with job creation. The establishment or expansion of businesses has highly variable effects on jobs. Obviously, some new firms add substantially to the job pool, but others produce only very modest gains in employment. Similarly, expansion of existing firms may lead to varying levels of new employment, or simply to changes in capital equipment and facilities.

Relocations and Displacement

Many critics of enterprise zones postulate that if zones are successful in attracting new investment and creating jobs, such growth will come at the expense of other urban areas. Thus, the argument goes, enterprise zones are a zero-sum strategy by which some areas gain jobs and business, while other localities lose out. Development decisions by firms simply are skewed toward enterprise zones rather than other viable sites.

As the discussion of incentives suggests, when firms assess enterprise zones against similar sites, they have a marginal comparative advantage, particularly for new business start-ups and decisions about new branch locations. Although critics regard this bias as a negative effect, zone advocates endorse it for targeting development to otherwise declining areas.

In addition to site displacement, critics see firms' relocations as the source for much of the growth within zones. These predictions have proven to be somewhat exaggerated. The study findings presented earlier show that relocating firms accounted for anywhere from 6 percent to 33 percent of zone business. It is not entirely clear from zone studies why such variations occur. A possible determinant may be the antirelocation provisions in some state zone programs (e.g. Ohio, Indiana) that restrict eligibility for incentives to new start-ups or existing firms. More importantly, relocating firms generally play only a small role in job creation and investment growth. Multi-case studies across zones consistently show that most new investment and employment in enterprise zones is from new business start-ups and expansion by existing firms. Thus, enterprise zones do not appear to create significant dislocation of economic activity.

Studies of the Costs of Enterprise Zone Programs

While analyses of job and investment impacts have increased, studies of program costs have remained rare. Several methodological barriers impede such efforts. Most state enterprise zone programs do not require detailed reports of annual program costs. Information on the amount of state revenue forgone through firms benefiting from zone incentives is usually unavailable because of the confidential nature of corporate tax returns. Thus, only a few studies have attempted to assess the costs of zone programs.

Studies of Costs for the Indiana Program

Wilder and Rubin (1988) examined the costs and benefits of the enterprise zone program in Indiana. Using data submitted to the state program office by zone directors, they found significant job growth and business expansion in certain zones. However, for some of the zones, moderate private sector expansion was accompanied by very high public expenditures. Estimates of public-to-private cost ratios ranged from 1/160 in Fort Wayne to 1/0.4 in Gary. These figures suggest that the capacity of public dollars to leverage private investment can vary considerably.

These findings were substantiated by James Papke (1988, 1989) in two subsequent studies of the Indiana program. His analyses of the fiscal and economic impacts of the Indiana zones shed additional light on their cost effectiveness. Using data from official firm-level tax records, Papke found that in 1987, the cost-per-job for enterprise zones in Indiana was a modest $4,100 in forgone tax revenue. However, he noted a considerable amount of variation by zone in cost-per-job figures: a high of $13,531 for Gary and a low of $389 for Madison, Indiana.

In their study of the Evansville enterprise zone, Rubin and Wilder (1989) found that the program generated lower cost-per-job figures than most other economic development programs in the public sector. The annual cost per new job in the zone was $1,045. When adjusted for those jobs created solely because of the existence of the enterprise zone, this figure was $1,372 per job. The analysis also revealed that new firms and businesses with 25 or more employees were the most cost efficient in creating jobs.

A more recent cost-benefit analysis of Indiana's zone program by Rubin, Brooks, and Buxbaum (1992) suggests that it has been cost efficient. To improve upon previous evaluations of the state program's costs, an expanded data base was created, using information from Indiana's Tax Commission, Department of Employment, Commerce Department, and various zone participants. Program costs were determined by calculating forgone state and local tax revenues, as well as administrative expenses. Total program costs were estimated at $20.6 million for 1990.[20] Approximately 85 percent of these costs were attributed to the program's inventory tax credit. Program benefits were estimated using an input-output model of the Indiana economy.[21] The primary direct benefit was state and local tax revenues, which totalled $272 million. The study estimated that for each $1.00 forgone by the inventory tax credit, local governments received an average of $2.20 in tax revenues. Despite the generally favorable outcomes, the researchers argued that greater cost efficiency could be attained by linking program incentives more directly to employment and investment objectives. For example, tax credits could be more directly tied to targeted training and hiring of zone residents, and/or to continued investment by zone businesses.

New Jersey Program Cost Analysis

An assessment of program costs in New Jersey by Rubin and Armstrong (1989) showed that approximately $51.6 million had been expended through tax incentives and direct expenditures for zone administration. More than 80 percent of the forgone revenue came from the program's sales tax exemption; few firms took advantage of the employee-related incentives. In a further test of the effects of enterprise zones, Rubin and Armstrong compared their costs and benefits under different scenarios: the best case scenario assumed that all firms were significantly influenced by the program; the worst case scenario assumed an influence only on those firms that had reported zone incentives to be their only or primary reason for business decisions. Using this methodology, the authors found that cost-per-job estimates ranged from $5,613 in the best case scenario to $13,070 in the worst case.

Interpretation of Cost Studies

The results to date show that aggregate costs incurred by states and localities for zone incentives range from less than $400,000 to over $50 million. The scale of these costs cannot be evaluated in isolation from the associated benefits. Unfortunately, though, most studies have not calculated cost/benefit ratios or cost-per-job figures. The most important point here is that the lack of comprehensive studies of program costs means that enterprise zones have not been subjected to the close fiscal scrutiny given other economic development programs. Important questions about the relative costs and benefits of enterprise zones remain unanswered. One question concerns the degree to which zones are cost efficient. Given the variability in their employment and investment levels, as well as in the targeting of new jobs to zone residents, a more detailed examination of the relative costs and benefits of enterprise zones is called for. Aggregate assessments of program costs and benefits are insufficient in that they do not provide the detailed analysis that could identify the most efficient zone contexts and program designs.

Moreover, although cost-benefit studies are important in assessing the efficacy of enterprise zones, they do not measure indirect impacts or costs. For example, each firm that relocates into a zone imposes an external cost on its previous community, in the form of lost income and taxes. Equally important are the opportunity costs of enterprise zone programs. The forgone revenue may represent a net loss of resources that could have supported other community-oriented programs. However, the revenue is only truly forgone if the development would have occurred in the absence of the enterprise zone. This perplexity is perhaps the most difficult one to resolve. Studies reviewed here suggest that investment decisions of existing firms are the most heavily affected by zone incentives. Thus an argument can be made that this particular form of new investment probably would not have occurred in the absence of the zone program. But what about the program costs for incentives to new or relocating firms? According to most of the literature on location decisions, these firms would not have been heavily influenced by the incentives, except "at the margin," and thus might well have made their investments without incentives. If so, those incentives were not cost-efficient.

Obviously, more research is needed to differentiate the components of program costs for new, existing, and relocating firms. In addition, these differentials should be compared to the relative direct benefits (e.g., investment, jobs) generated by each category. If enterprise zones prove to be net revenue enhancers, then an argument can be made that program gains might actually increase the fiscal ability of localities and states to address other pressing urban problems. It is exceedingly important that states conserve limited resources by monitoring the actual and indirect costs of zone programs, and by redistributing any net gains in revenue to other areas of need.

Planning and Policy Implications

While states share common program objectives of economic development and job creation, the formulation and implementation of enterprise zone programs vary extensively. Even the definition of what constitutes an enterprise zone varies by state. Moreover, as development instruments, enterprise zones are distinguished from other economic development approaches not by a new method of revitalization, but rather by the packaging and geographic targeting of traditional development incentives and tools. Since most states offer similar forms of development incentives, a question arises as to the influence of program design on program effectiveness. The most recent research on enterprise zones has attempted to distinguish the program and nonprogram factors associated with job and investment growth.

As suggested in the discussion above, the key program attribute—development incentives—can have several roles in stimulating economic activity. However, other factors have been identified as having more consistent effects on economic growth within the zones. Research reviewed here suggests that areas that have suffered severe economic blight require much more assistance than enterprise zone programs can provide. The enterprise zone is much more effective if it incorporates an area that still retains a stable economic base. Moreover, zones with a significant amount of industrial and/or commercial land use have more propensity to revive under zone designation. Finally, autonomous management of enterprise zone activities seems to be the most effective.

Analyses of program structure (e.g., Green and Brintnall 1987; Erickson et al. 1989) suggest that those state programs are most successful that: (1) restrict the number of enterprise zones, (2) use a competitive designation process, and (3) provide a wide array of development incentives. The selection process of state programs evaluates the potential comparative advantage of zone areas, and can target the areas that have developed strong local support. Thus, state programs should use zone designation sparingly, and should convey zone status as part of a conscious strategy to confer comparative advantage on certain localities. This more conservative approach to zone programs will also help to contain costs and will allow closer monitoring and evaluation.

Zone programs must also develop more specific employment goals. Currently, most programs have very broadly stated goals of "increasing employment opportunities." Seldom is there a specific goal of targeting those opportunities to particular persons or groups within the zone. If zones are to effectively and efficiently improve the employment and incomes of disadvantaged residents, a more direct link must be forged between that objective and the design and implementation of the zones. Such a link will also require more emphasis on job training and educational services to match the skills of zone residents with the requirements of zone firms. If zones are treated as a "metropolitan" employment strategy, policymakers must recognize the resulting leakage of zone benefits, and address the obvious employment disadvantage that zone residents experience.

Finally, urban revitalization is most effectively addressed through a strategic planning process that encompasses the entire urban area and includes enterprise zones as one aspect of a development strategy. Moreover, since zone management, program coordination, specific employment goals, and community organization aspects are important to the success of individual enterprise zones, zone organizations should also be engaged in strategic development planning. Such a planning process should both identify development goals and objectives for the enterprise zone, and establish strategies for overcoming its limitations.

From a planning perspective, it is disheartening to note that not one of the studies of state enterprise zones discussed above found that a planning process was a characteristic (important or otherwise) of the evolution of enterprise zones. This is true whether for zones serving as elements of larger-scale development plans, or for zone organizations creating their own development plans. The failure to meld a planning process with enterprise zone activity may be due, in part, to state programs having evolved before strategic planning was recognized as an important activity for the public sector. This issue has, in fact, been addressed for federal enterprise communities and empowerment zones. The Clinton Administration required that all applicants engage in strategic planning to decide how the goals of the federal legislation would be addressed. It will be interesting to see how this aspect of the federally designated zones fares, and whether this element of enterprise zone performance will be transferred to state zones.

Conclusion

The content of this review makes clear that the policy debate over enterprise zones must shift from its rhetorical basis to one more firmly grounded in empirical data. Additional research is needed to monitor the effects of enterprise zones as they mature. One of the most important questions that remains unanswered is whether enterprise zones are more or less cost effective than alternative initiatives for community revitalization. The analytic results shown here demonstrate that enterprise zone programs must be viewed as simply one among a number of tools that can be used to address the needs of urban communities. The revitalization of declining urban areas requires a comprehensive, coordinated, and planned set of strategies aimed at increasing economic opportunities for residents, as well as for business owners.

NOTES

1. The enactment of the Federal enterprise zone initiative under the Housing and Community Development Act of 1987 was a hollow victory for zone advocates. Although the Act allowed HUD to designate up to 100 Federal enterprise zones, it provided no real fiscal incentives. No designations ever took place.
2. Under the program, urban empowerment zones have been designated in Atlanta, Baltimore, Chicago, Detroit, New York, and Philadelphia-Camden. Supplemental empowerment zones have been designated in Los Angeles and Cleveland. Four cities have "enhanced" enterprise communities—Boston, Houston, Kansas City, and Oakland. The remaining empowerment zones/enterprise communities are found in rural areas and other urban communities.
3. The original concept of the enterprise zone envisioned by Peter Hall (1977) seeks to generate new investment and employment in declining subareas of cities through targeted fiscal incentives and regulatory relief. The concept embodies the basic assumptions of supply-side economics, that is, that by reducing constraints on the exercise of free enterprise (e.g., governmental "red tape") and minimizing the risk and cost of private sector investment, new development will occur that generates additional employment and public revenue.
4. Organizations such as the U.S. Conference of Mayors, the National Governors' Association, and the U.S. Chamber of Commerce have supported the zone concept on behalf of their respective members (U. S. Congress, Hearings, 1981, 1991, 1992).
5. Some critics argue that the jobs created by zone businesses are much more likely to be low-paying and provide little job security (Clarke 1982, Goldsmith 1982). Representatives of the AFL-CIO and the International Ladies' Garment Workers' Union have warned that enterprise zones will attract labor-intensive industries (e.g., apparel manufacturing) that are foot-loose and seek low-wage, nonunion workers (U.S. Congress, House Committee on Ways and Means, Hearings, 1991).
6. The HUD study sites included: Bridgeport, CT; Chicago, IL; Dayton, OH; Louisville, KY; Macon, MO; St. Louis, MO; Michigan City, IN; Tampa, FL; Thief River Falls, MN; and York, PA.
7. In terms of general attributes, the study found that most zone firms were either new (38 percent) or existing (34 percent) businesses. Approximately 13 percent of zone firms were branch expansions of businesses located outside the zone. Another 15 percent of zone firms had relocated from sites outside the zone areas.
8. The California zone program takes two forms: (1) a traditional zone program and (2) a special employment and economic incentive program. Both programs provide tax credits for hiring unemployed residents, sales tax credits, and accelerated depreciation on business property. Under the first program, any firm located within a zone can receive the tax breaks. The second program allows any firm to become "certified" for tax benefits by employing specified percentages of residents from "high density unemployment areas."
9. Porterville had no net change, and Bakersfield and Yuba City experienced modest net losses of 4 and 5 establishments, respectively.
10. This method for determining the economic impact of a zone was pioneered by Rubin and Wilder (1989) in a detailed study of the Evansville, Indiana enterprise zone.
11. Los Angeles-Watts, Fresno, Agua Mansa, San Diego-Barrio Logan, and Los Angeles-Central.
12. New employment was heavily concentrated in four industrial sectors: manufacturing (3,177), trade (2,102), finance/ insurance /real estate (1,942) and services (1,198). New investment was focused on finance/insurance/real estate ($438 million), manufacturing ($152 million), and trade ($143 million).
13. The initial cohort included zones in the Bronx, Brook-

lyn, Gloversville, Islip, Lackawanna, Ogdensburg, Olean, Queens, Troy, and Syracuse.

14. Other significant factors included: labor force attributes, access to transportation, availability of low interest loans/grants, and quality of public services.
15. Although the majority of zone businesses (69 percent) existed prior to zone designation, approximately 83 percent of new jobs were generated by the establishment of new firms. In terms of firm size, the study showed that over 64 percent of new jobs were created in firms having 50 or more employees. In fact, job growth in the Evansville zone was dominated by branch operations of large firms with nonlocal headquarters.
16. The overwhelming majority of new jobs were generated by businesses involved in transportation, trade, and distribution activities. The authors attributed this outcome to the relative attractiveness of inventory tax credits enjoyed by such firms.
17. These findings are based on a sub-group of 213 zones for which complete data were available.
18. High performance zones were those in which: a) ten or more firms per year had made investments, or b) 150 jobs were created or retained per year.
19 Cost breakdowns revealed that forgone revenues were $2.5 million for the State, and $17.6 million for localities.
20. Program benefits included state/local tax revenues, employee wages, the value of produced goods/services, and investment in business facilities/equipment.
21. Although superficially a new incentive exists in the form of regulatory relief, in practice this element is used in a very narrow, restricted manner. (See Sheldon and Elling 1989, Green and Brintnall 1987.)

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APPENDIX. Summary of Enterprise Zone Studies

Author(s) and Year	Geographic Area (Study Time Period)	Methodology	Key Findings
Jones (1985)	Bridgeport, CT (1980–84)	Municipal records used to assess building permit and property transfer activity in the zone and a non-zone area; Wilcoxon signed rank test used to test significance of differences.	*No significant difference found in activity levels within the zone after designation *Non-zone area had higher levels of development activity than zone did.
Jones (1987)	Decatur, IL (1981–85)	Building permit and property transfer activity examined for the zone before and after designation; Chi-square analysis used to test significance of differences.	*Statistically significant increases occurred in the volume and value of building activity within the zone, particularly for industrial/commercial activities. *Increased development in zone attributed to national growth trends rather than program
Connecticut Dept. Econ. Dev. (1985)	Connecticut (6 zones) (1982–85)	Descriptive analysis of program impacts using data from biannual surveys of local zone managers; summary data presented on job/investment changes by zone and type of economic activity.	*4,300 new jobs and 4,200 retained positions attributed to zone program *Over $113 million in new investment found in zones since their designations *Majority of new investments made by existing firms, particularly commercial, retail, and mixed use activities
HUD (1986)	Bridgeport, Chicago, Dayton, Louisville, St. Louis, Tampa, Michigan City, IN; Macon, MO; Thief River Falls, MN; York, PA. (program start-up dates to 1985)	Descriptive analysis of program attributes and impacts; detailed case studies based on about 200 field and phone interviews with program administrators, business owners, and zone residents.	*Over $147 million and 6,730 new jobs reported *New and expanded firms accounted for majority of job growth. *Small firms (<50 workers) were responsible for majority of job growth. *Zone incentives affected expansion of existing firms, but were less influential in the locational decisions of new firms.
Funkhouser and Lorenz (1987)	Maryland (10 zones) (1983–85)	Descriptive analysis of costs associated with first two years of EZ program. Costs included program administration, infrastructure improvements, tax credits, job training, and loans/grants. Data sources: state tax records, annual program reports. Results reported by zone and SIC.	*Zone firms were making minimal use of tax incentives. *Estimated cost per job generated was $1,400 or $57,000, depending on data source used. *Findings attributed to newness of program and usual lags in applications for credits and reporting
Brintnall and Green (1988)	17 states (program start-up dates to 1985)	Program materials and results of mail survey of program officials used to develop indices for (1) administrative oversight of program, and (2) degree of private sector influence. Index measures used to classify state programs.	*Four categories of state programs identified: managed, activist, hands-off and private *Activist state programs (i.e., with high levels of program management and private sector involvement) were more innovative and risk averse (e.g. Illinois, California, Indiana, Pennsylvania).
California Auditor General (1988)	California (10 zones/3 incentive areas) (1986–87)	Descriptive analysis of initial program impacts. Data drawn from state records and telephone survey of zone firms.	*3,600 new jobs reported by firms in zones/incentive areas *Net employment increase of 1,800 jobs, 1,150 of which were obtained by economically disadvantaged persons *2,200 new firms established, resulting in net increase of 79 businesses

APPENDIX continued

Author(s) and Year	Geographic Area (Study Time Period)	Methodology	Key Findings
U.S. General Accounting Office (1988, 1989)	Cumberland, Hagerstown and Salisbury, MD (1980–87)	Descriptive report of job growth and business attitudes towards incentives Data taken from monthly state unemployment records and a special survey of firms in the three zones (N=493 firms). Interrupted time series analysis (ARIMA modeling) used to assess changes in employment.	*Employment growth reported in each zone but not attributed to zone program *Majority of firms participating in program were existing businesses. *Most firms did not regard program incentives as important in making locational decisions.
Nelson/Whelan (1988)	Louisiana (1983–85)	Mail survey of zone firms used to assess program impacts and attitudes of business owners towards incentives (N=19 firms)	*$27.4 million in new investment and 1,036 new jobs reported by survey respondents *Incentives were viewed as important for 50% of respondents when making locational decisions, but far less critical for expansion and investment decisions.
Papke (1988)	Indiana (1986)	Economic analysis of program impacts Data taken from annual EZ firm registration/survey forms (N=1054) Wages, tax credit costs, etc. for each zone Econometric model used to estimate effects of incentives on profitability rates of zone firms	*5,728 new jobs created in 1986, 11% filled by zone residents *94% of tax incentive costs accounted for by inventory tax ($11.6 million in 1986) *Tax credits increased net profit rates by 1–7%, depending on location and type of business.
Papke (1989)	Indiana (1987)	Descriptive analysis of job generation and tax credit usage by zone and type of firm Data drawn from annual business registration/survey forms (N=1068)	*2,906 new jobs created in 1987, 17% filled by zone residents *89% of tax incentive costs accounted for by inventory tax credits ($10.7 million in 1987) *Average cost per new job was $4,100. *Majority of new jobs created by manufacturing firms *Few firms took advantage of wage-related tax credits.
Staley (1988)	Dayton, Ohio (2 zones) (1983–87)	Case study of program impacts and attitudes of business owners Data obtained from survey of zones' businesses (N=33 of 41) and municipal records.	*878 new jobs created within zones, 26% filled by zone residents *Majority of new jobs were for unskilled labor. *Incentives viewed as marginal factors in locational decisions, but critical to expansion decisions of existing firms
Wilder/Rubin, B. (1988)	10 zones, with emphasis on Evansville, IN (1983–86)	Descriptive analysis of job and investment impacts using data from quarterly zone reports and detailed case study	*6,600 new jobs and $295 million in new investment reported in the 10 zones; highly variable outcomes across zones *Over 1,600 new jobs and 47 new businesses reported in Evansville *Success in Evansville attributed to industrial land use mix, effective program management, and strong local support

APPENDIX continued

Author(s) and Year	Geographic Area (Study Time Period)	Methodology	Key Findings
Erickson et al. (1989)	357 zones in 17 states (start-up dates to 1986)	Statistical analysis (via correlation/ regression models) of relationship between employment/investment and program attributes Data taken from HUD survey of zone coordinators (N=357 zones, N=2014 businesses, N=186 communities) Mostly averages reported.	*An average of 333 new jobs and $23.4 million in new investment reported per zone *81% of firms making investments were new businesses or expansions. *New investment and employment was dominated by manufacturing firms (73% of total impacts). *Competitive/restrictive designation policies and multiple forms of incentives were associated with greater program success.
Rubin, B./Wilder (1989)	Evansville, IN (1983–86)	Follow-up to 1988 case study Shift-share analysis used to isolate effects of zone from areal and industry-related growth Data obtained from local program database and state tax credit reports	*76% of job growth due to zone's comparative advantage *Average annual cost/new job was $1,372. *Services, transportation/construction industries most cost-effective as job generators *No direct relation shown between size of tax savings and job generation
Rubin, M./ Armstrong (1989)	New Jersey, 10 zones (1987–89)	Economic analysis (input-output) of program costs (administrative expenses and incentives) and benefits (jobs, income, corporate/personal taxes) Data from mail survey of zone firms (N= 478) and records of state commerce and tax departments	*9,193 new jobs reported for 1985–88 period *$803 million in new investment reported for 1987 *For 87–88 period, total direct program costs were $51.6 million; total direct benefits ranged from $39 to $107 million depending on assumed relationships. *Costs per job for 87–88 ranged from $5,613 to $13,070. *Majority of firms (70%) report that incentives influenced locational/expansion decisions. *Larger firms more prone to value program benefits *Sales tax exemption more valued than business tax credits or unemployment tax rebates
Sheldon/Elling (1989)	Illinois, Ohio, Indiana, Kentucky (47 zones) (start-ups to 1988)	Comparative analysis of business and employment growth Regression analysis used to assess influence of program factors on variations in zone outcomes Data obtained from telephone and mail survey of state/zone administrators	*An average of 499 new jobs reported per zone *Zones varied significantly between and within states in terms of job generation. *Zone success most strongly associated with degree of program management, array of services/incentives, differences in state/local economic conditions *Indiana was most successful state program.

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Author(s) and Year	Geographic Area (Study Time Period)	Methodology	Key Findings
Hamilton Associates (1990)	(1988–90)	Economic analysis of costs and benefits and case studies of 4 zones Data drawn from survey of zone businesses (N=163) and administrators (N=19), and records of state departments of labor, economic development and taxation	*About 1,200 new jobs were created in zones during 1988–89. *Brooklyn and Syracuse zones led employment growth with 500 and 300 new jobs, respectively. *54% of zone firms considered program incentives to be important factors in investment and locational decisions. *Cost benefit estimates ranged from $15.8 million net benefit (best case scenario) to $6.1 million net costs (worst case); break-even scenario was deemed easily attainable over 10 year period. *Annual cost/job over 10 years was $4283 (worst case).
Litster (1990)	California (86–88)	Descriptive analysis of employment impacts and influence of incentives Data obtained from survey of business owners across zones (N=137)	*2,518 new jobs created in zones *54% of employment gain generated by new firms *Majority (55%) of business owners did not regard incentives as important in making locational decisions.
Rubin, M., et al. (1992)	Indiana (12 zones) (1989–90)	Economic analysis (input-output) of program costs and benefits Data obtained from annual business registration forms, records from state departments of employment and taxation, and interviews with local zone managers and business owners (N=1,964 businesses)	*2,024 new jobs reported by zone firms for '89–90 *Large firms (>100 workers) accounted for 62% of new jobs and 68% of new investment. *For 1990, total direct program costs were $20.6 million; total direct benefits were $199 million in investment and $272 million in new state/local tax revenues.
Dowall, D., et al. (1994)	California 10 zones, 3 incentive areas (1986–90)	Descriptive and economic analyses (shift-share) of employment impacts Data drawn from County Business Patterns, U.S. Census reports, and survey of zone administrators and businesses (N=159 firms)	*Total employment across zones increased by 13% (16,864 jobs) between 1986 and 1990. *Large firms generated 54% of employment growth, medium-sized firms-33%, and small firms-13%. *No causal link found between employment changes and zone program *Only the San Jose and Los Angeles-Pacoima zones had experienced a comparative advantage in terms of economic growth.